

Survival-Informed Multi-Omics Kernel Fusion for Cancer Subtyping

Xiaojuan Ding , Pengcheng Shi , Xin Wang , and Hui Cao

Abstract—Cancer molecular heterogeneity impedes precise subtyping and personalized therapy. Current multi-omics integration methods often overlook clinical relevance and kernel redundancy, yielding subtypes with limited prognostic utility. Here, we introduce Survival-Informed Multi-omics Kernel Fusion (SIMKF), a framework that synergizes survival-guided kernel selection with distribution-aware fusion to uncover clinically distinct subtypes. SIMKF addresses the limitations of current multi-omics integration methods by combining survival-guided kernel selection, adaptive weighting based on maximum mean discrepancy, and spectral clustering to integrate survival information with multi-omics data. This approach significantly outperforms existing techniques across five TCGA cancer datasets. Notably, in breast cancer, it successfully identifies five distinct subtypes with pronounced survival differences, revealing a nonlinear relationship between methylation levels (hypermethylation correlating with better prognosis, hypomethylation with poorer outcomes) and survival outcomes, while aligning closely with established clinical subtypes. As an automated, tuning-free tool for precision oncology, SIMKF not only uncovers prognostic biological mechanisms but also translates directly into clinically applicable subtyping models.

Index Terms—Multi-omics integration, cancer subtyping, survival analysis, kernel fusion, multiple kernel learning.

I. INTRODUCTION

IN RECENT years, large-scale international initiatives such as The Cancer Genome Atlas (TCGA) [1] have enabled the accumulation of multi-dimensional omics data—including genomics, transcriptomics, epigenomics, and proteomics—offering unprecedented opportunities to dissect tumor heterogeneity. However, traditional molecular subtyping approaches based on a single omics layer (e.g., gene expression profiles) often fail to capture the dynamic interplay among biochemical processes within the tumor microenvironment. This limitation

has prompted researchers to adopt multi-omics integration methods to achieve more robust and biologically meaningful subtyping results through coordinated data mining. Nevertheless, several critical challenges remain in practical applications [2]: (1) the high-dimensionality and limited sample size problem; (2) heterogeneity across data modalities, with different omics layers exhibiting diverse statistical properties and biological implications; and (3) the complexity of modeling molecular interaction networks, which requires effective integration of cross-omics systems-level information.

To address these issues, current multi-omics integration strategies have evolved along three major technical pathways [3]. Early fusion approaches, such as LRAcluster [4], concatenate multi-omics feature matrices for joint analysis. While conceptually simple, they are vulnerable to issues of data heterogeneity and the curse of dimensionality. Late fusion methods, such as PINS [5], follow a divide-and-conquer strategy by clustering each omics type independently and then aggregating the results, improving flexibility but risking the loss of weak signals. Intermediate fusion approaches strive to balance information integration efficiency and model robustness through innovative algorithmic frameworks. Representative methods include similarity-based fusion such as SNF [6], joint dimensionality reduction models like JIVE [7], and probabilistic graphical models exemplified by iCluster [8]. More recently, Multiple Kernel Learning (MKL) has emerged as a promising direction within the intermediate fusion category. MKL-based methods construct a separate kernel matrix for each omics type and perform kernel combination and learning within a unified framework, thereby capturing the nonlinear relationships and complementary information among data modalities [9]. Compared to traditional fusion strategies, MKL offers greater flexibility and expressive power. A notable example is the CIMLR algorithm proposed by Ramazzotti et al. [10], which constructs multiple Gaussian kernels for each omics type and fuses them into a unified similarity matrix.

To enable more precise cancer subtyping and uncover key molecular mechanisms, this study is driven by the following two core motivations.

Q1: The disconnect between clinical prognostic information and molecular feature mining. Traditional methods in integrating multi-omics data often rely on manual selection of kernel functions or preset fixed clustering numbers, leading the fusion process into clinically non-informative partitions—where the generated subtypes may be mathematically separable but lack prognostic

Received 9 July 2025; revised 5 December 2025; accepted 28 December 2025. Date of publication 5 January 2026; date of current version 3 April 2026. This work was supported in part by the Project of Philosophy and Social Science Research in Colleges and Universities in Jiangsu Province under Grant 2024SJYB0220 and in part by the Scientific Research Innovation Project of Nanjing University of Finance and Economics under Grant XKYC2202507. (Corresponding author: Xiaojuan Ding.)

Xiaojuan Ding, Pengcheng Shi, and Xin Wang are with the School of Computer and Artificial Intelligence, Nanjing University of Finance and Economics, Nanjing 210023, China (e-mail: wjswsl@163.com).

Hui Cao is with the PetroChina Changqing Oilfield Company, No. 1 Oil Production Plant, Changqing Oilfield Yan'an, Shaanxi 716000, China.

The code and datasets are available at <https://github.com/XiaojuanDing/TCBB-SIMKF>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TCBBIO.2026.3650772>, provided by the authors.

Digital Object Identifier 10.1109/TCBBIO.2026.3650772

discriminative power due to the absence of clinical priors such as survival information (as evidenced by overlapping KM curves and insignificant log-rank tests).

- Q2: The inherent heterogeneity of multi-omics data poses challenges in modeling cross-modal molecular relationships, severely limiting the precise identification of cancer prognostic subtypes. Current multiple kernel clustering methods (e.g., MKKM) face two major biological limitations when integrating different omics (e.g., genomics, epigenomics, transcriptomics): loss of critical survival signals and lack of clinical interpretability.

To address the above challenges, this paper proposes a method called Survival-Informed Multi-Omics Kernel Fusion for Cancer Subtyping (SIMKF). Building upon the traditional multi-omics clustering framework, SIMKF introduces a kernel selection mechanism for the generated initial kernel set that integrates prior knowledge with diversity evaluation. Specifically, multiple base kernels are generated for each omics view, and a subset of kernels with greater discriminative power and complementarity is selected by considering their correlation with disease-related phenotypes (such as survival information, prognostic signals, etc.) and the diversity metrics between the kernel functions. Subsequently, the selected kernels are assigned weights based on the Maximum Mean Discrepancy (MMD) criterion, effectively reflecting the importance of different kernels in capturing modal feature structures and avoiding uncertainties caused by manually setting hyperparameters. Finally, the weighted fusion kernel similarity matrix is input into the clustering module to complete the subtype identification of samples. This method not only enhances the robustness and interpretability of multi-omics data fusion but also significantly improves clustering performance and survival stratification.

The contributions of this paper can be summarized as follows:

- This study introduces a kernel selection strategy combining clinical survival data and kernel diversity to automatically identify optimal multi-omics kernels and determine the most discriminative subtypes.
- To address the issue of traditional weighting methods obscuring critical survival signals, this study proposes an unsupervised weighting mechanism based on omics distribution differences.
- Our method outperforms nine state-of-the-art approaches in identifying clinically relevant subtypes with significant survival differences across five multi-omics datasets.

II. RELATED WORK

Multi-omics clustering algorithms are generally categorized into early, intermediate, and late integration strategies based on the fusion stage.

Early integration methods concatenate omics matrices into one composite matrix, followed by conventional clustering algorithms like K-means or hierarchical clustering. Though easy to implement, they are prone to data heterogeneity and high dimensionality. For example, Zhang et al. [11] used Cox regression to select prognosis-related features and applied super

K-means clustering, while Wu et al. [3] jointly optimized omics data using a low-rank parameter matrix.

Late integration clusters each omics modality separately and then combines results using strategies like consensus clustering. While offering flexibility, these methods may overlook weak signals. Sun et al. [12] built neural networks for each modality and linearly combined their outputs. Wang et al. [13] proposed a multi-view network consensus model, and Hoadley et al. [14] used indicator matrices for integration.

Intermediate integration balances fusion depth and model robustness by leveraging architectural designs to integrate multi-omics features. Representative examples include JIVE, based on joint dimensionality reduction [6], and iCluster, which uses probabilistic graphical models [7].

Recently, multiple kernel learning (MKL) has emerged as a powerful intermediate integration tool due to its ability to model data from different perspectives [15], [16]. MKL applies various kernels to capture complementary structures across omics. SNF [5] builds patient similarity networks using scaled exponential kernels and integrates them via message passing. Feng et al. [17] used Gaussian kernels to generate similarity matrices and fused them into a global kernel. However, these methods assume equal contribution from each modality, which may not reflect biological reality.

To model effect heterogeneity, Ramazzotti et al. [10] proposed CIMLR, which constructs multiple Gaussian kernels for each omics type and combines them to form a fused similarity matrix. It introduces rank constraints and graph diffusion to refine similarity learning. However, CIMLR does not explicitly model the heterogeneous contributions of each modality, potentially reducing clustering effectiveness [18].

To address these limitations, enhanced methods have been proposed. John et al. [19] introduced a density-aware kernel enhancing similarity among local neighbors. Chen et al. [20] developed SFHOP, which builds high-order proximity graphs per modality and fuses them into a unified structure. Yang et al. [18] proposed GA-fKPLS, which uses genetic algorithms to jointly optimize kernel parameters and weights. Wei et al. [21] introduced hMKL, a hierarchical framework that separately optimizes kernel parameters and fusion weights. Yet, these methods often require manual hyperparameter tuning, which reduces stability and generalizability in unsupervised settings.

Moreover, many approaches neglect prior biological knowledge, relying mostly on highly variable features that may capture general variation rather than prognostic relevance [22]. However, factors such as cell-of-origin [23], histological subtype, tumor microenvironment [24], dedifferentiation [25], and signaling pathways [26] play crucial roles in prognosis. Incorporating clinical data like survival can thus enhance subtyping accuracy.

To capture prognostic signals, Arora et al. [27] proposed survClust, which uses survival-weighted distance matrices to suppress irrelevant features and identify subtypes with significant survival differences. Despite its benefits, survClust averages distances across omics and involves computationally intensive steps like cross-validation and label alignment. To reduce such overhead, Cao et al. [28] proposed wMKL, integrating survival relevance into Gaussian kernel construction. However, wMKL's

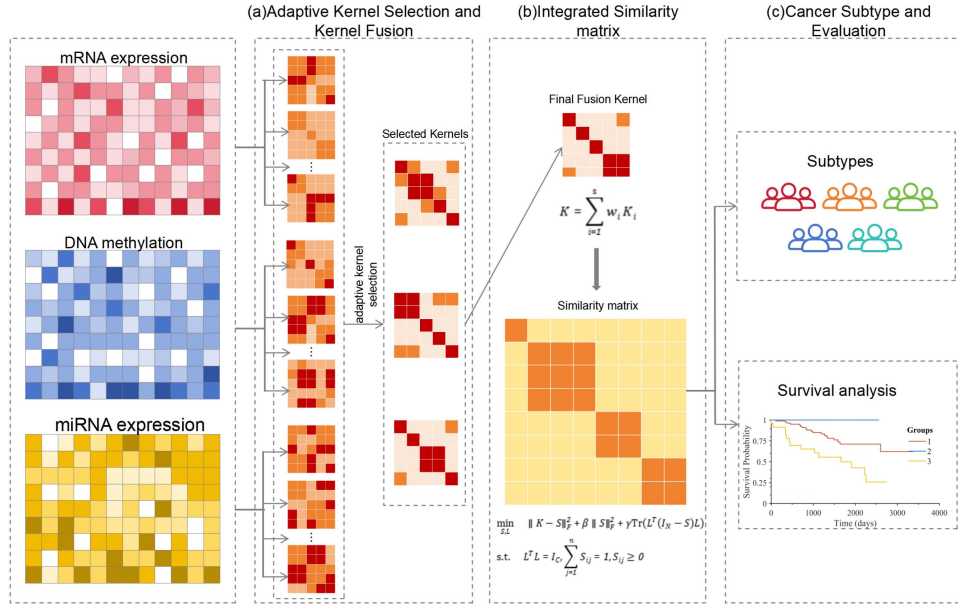


Fig. 1. The flowchart of Survival-Informed Multi-Omics Kernel Fusion for Cancer Subtyping (SIMKF). SIMKF mainly includes three steps: (a) Adaptive Kernel Selection and Kernel Fusion, (b) Integrated Similarity matrix, and (c) Cancer Subtype and Evaluation.

single-kernel structure limits its flexibility in capturing structural heterogeneity.

Unlike CIMLR and SNF, which either assign equal weights or iteratively learn kernel weights without incorporating clinical priors, our method explicitly integrates survival-guided kernel selection and structural diversity to enhance biological interpretability. Furthermore, instead of relying on diffusion-based fusion as in SNF, we employ an MMD-based weighting mechanism that quantifies the distributional distance between each kernel and the fusion target. This design enables interpretable, efficient, and iteration-free weight learning, improving clustering robustness and subtype identification accuracy.

Our approach differs by jointly selecting base kernels with high phenotypic relevance and structural diversity, enhancing clinical interpretation while filtering noisy kernels. To avoid reliance on iterative tuning, we introduce an MMD-based weighting mechanism that calculates the distributional distance between each kernel and the fusion target. This enables interpretable, efficient, and iteration-free weight learning, improving clustering robustness and subtyping performance.

III. METHOD

In this section, we provide a detailed description of our proposed framework, which is illustrated in Fig. 1. The method mainly includes three steps: (a) Adaptive Kernel Selection and Kernel Fusion, (b) Integrated Similarity matrix, and (c) Cancer Subtype and Evaluation. The overall pipeline is designed to achieve both effective integration of heterogeneous omics data and accurate identification of clinically relevant cancer subtypes. To facilitate understanding of the mathematical formulation and symbols used throughout this section, a summary of all key notations is provided in Table I.

TABLE I
SUMMARY OF NOTATIONS.

Notation	Meanings
n	Number of samples
d_i	Number of features in the i -th omics
m	Number of candidate kernels per omics type
C	Number of clusters
w_i	Weight assigned to the i -th kernel function
β	The regularization parameters of the norm for S
γ	The weight parameter on the low-rank constraint
$\ \cdot\ _F$	Frobenius norm, used to measure matrix reconstruction error
$\text{Tr}(\cdot)$	The trace of a matrix
$\mathbf{X}_i \in \mathbb{R}^{d_i \times n}$	Data matrix of the i -th omics type
$\mathbf{K}_i^j \in \mathbb{R}^{N \times N}$	The j -th kernel matrix of the i -th omics
$\mathbf{K} \in \mathbb{R}^{N \times N}$	Fused kernel matrix
$\mathbf{S} \in \mathbb{R}^{N \times N}$	Learned structure-preserving similarity matrix
$\mathbf{L} \in \mathbb{R}^{N \times C}$	Low-dimensional embedding matrix representing N samples in C -dimensional space
$\mathbf{I}_N$	N -dimensional identity matrix

A. Adaptive Multi-Omics Kernel Selection Strategy

Given a multi-omics dataset $X = \{X_1, X_2, X_3\}$, for any view $X_i = [x_{i1}, x_{i2}, \dots, x_{in}] \in \mathbb{R}^{d_i \times n}$ ($i = 1, 2, 3$), which respectively represent the data matrices of miRNA expression, mRNA expression, and DNA methylation. Each omics data needs to be processed by random smooth nucleation to generate candidate kernel sets. Therefore, we use the scaled exponential similarity kernel [29] as the seed kernel to perform kernel mapping for each view. The kernel similarity between sample x_i and sample x_j is defined as [30]:

$$\phi^x(x_i, x_j) = \begin{cases} \exp\left(-\frac{d(x_i, x_j)}{\omega \epsilon_{ij}}\right), & \text{if } x_i \in G_f(x_j) \text{ or } x_j \in G_f(x_i), \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

where $d(x_i, x_j)$ is the Euclidean distance between x_i and x_j , and $G_f(x_i)$ represents the set of f nearest neighbors of x_i , ω is a hyperparameter, ϵ_{ij} is a scaling factor. By randomly perturbing parameters ω and ϵ_{ij} , candidate kernels with high expected diversity can be generated. For each input omics data X_t , the above method is used to generate a candidate kernel set $\mathcal{S}_t = \{\mathbf{K}_1^t, \dots, \mathbf{K}_m^t\}$, where m is the number of basic kernels preset for that modality. Three candidate kernel sets corresponding to the three omics data are ultimately obtained: $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$.

The performance of each candidate kernel is then evaluated using survival information to incorporate clinical relevance. Specifically, for each kernel $\mathbf{K}_p$, k -means clustering is performed to obtain sample group labels, followed by a log-rank test to compute the p -value based on actual survival data. A smaller p -value indicates stronger association between the clustering structure and patient prognosis, reflecting greater disease relevance.

In addition to phenotype correlation, to avoid redundancy between kernel functions, we introduce a kernel diversity indicator to evaluate the complementarity of candidate kernels. Taking the miRNA expression omics as an example, the generated candidate kernel set is denoted as $\mathcal{S}_1 = \{\mathbf{K}_1^1, \dots, \mathbf{K}_m^1\}$. For any candidate kernel $\mathbf{K}_l^1$ ($l = 1, \dots, m$), through singular value decomposition (SVD), we obtain $\mathbf{K}_l^1 = \mathbf{U}\mathbf{S}\mathbf{V}^T$, where $\mathbf{S} = \text{diag}(\sigma_1, \dots, \sigma_n)$ with $\sigma_1 > \sigma_2 > \dots > \sigma_n$. If σ_1 is significantly larger than the other singular values, it indicates that the data is dominated by a single mode; if all singular values tend to be balanced, it reflects that the data contains multiple modes (i.e., diversity). According to the definition, singular values are the square roots of the eigenvalues of the kernel matrix. By calculating the eigenvalues of the kernel matrix $\lambda = \{\lambda_1, \dots, \lambda_n\}$, we can obtain the target singular values $\sigma = \{\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}\}$. Based on this, the kernel diversity score is defined as [31]:

$$\text{div} = -\log_n \left(\frac{\sqrt{\lambda_1}}{\sum_{i=1}^n \sqrt{\lambda_i}} \right), \quad 0 \leq \text{div} \leq 1. \quad (2)$$

Finally, to balance clinical relevance and diversity, an F -value-based criterion is employed to jointly assess each kernel. For every omics type, 55 candidate kernels with varying parameters are generated, and the one achieving the highest F -value is selected as the optimal base kernel $\mathbf{K}_i$ for subsequent multi-kernel fusion and cancer subtype identification.

B. Weight Learning and Multi-Kernel Similarity Fusion

To effectively fuse kernel functions with complementary structures but significant distribution differences across different modalities, this paper adopts the Maximum Mean Discrepancy (MMD) as the criterion for measuring distribution differences between kernel functions [32]. Specifically, given a filtered kernel set $\{\mathbf{K}_1, \mathbf{K}_2, \dots, \mathbf{K}_m\}$, the average pairwise MMD between each $\mathbf{K}_i$ and the remaining kernels is calculated to evaluate its representativeness and uniqueness. A larger MMD value indicates that the kernel has higher distribution dissimilarity, making it a more important source of information for fusion.

For each candidate kernel $\mathbf{K}_i$, its final weight w_i is calculated as:

$$w_i = \frac{\text{MMD}(\mathbf{K}_i)}{\sum_{j=1}^m \text{MMD}(\mathbf{K}_j)} \quad (3)$$

where $\text{MMD}(\mathbf{K}_i)$ denotes the average MMD of $\mathbf{K}_i$ with all other kernels. This normalization ensures that $\sum_{i=1}^m w_i = 1$, facilitating linear combination. Once the weights are obtained, a weighted fusion strategy is applied to construct the integrated similarity matrix $\mathbf{K}$:

$$\mathbf{K} = \sum_{i=1}^m w_i \mathbf{K}_i \quad (4)$$

This approach preserves the core information from each omics view while effectively suppressing redundancy, thereby enhancing the performance of downstream clustering tasks.

This novel kernel fusion mechanism adaptively assigns importance weights to multi-omics data kernels without complex iterative optimization. By directly quantifying distributional differences between omics-specific kernels in an unsupervised manner, it prioritizes kernels capturing clinically relevant patterns. Crucially, kernels are pre-filtered using survival information—only those significantly correlating with patient prognosis (e.g., via log-rank testing) enter the fusion stage. The weights then amplify kernels that provide complementary biological insights.

C. Optimization Algorithm

After completing kernel selection and weight fusion, this study further designs a unified optimization framework to enhance the expressiveness of the fused similarity matrix under clustering tasks, and guide the structural representation of samples in a low-dimensional space. The optimization model is formulated as follows:

$$\begin{aligned} \min_{\mathbf{S}, \mathbf{L}} \quad & \|\mathbf{K} - \mathbf{S}\|_F^2 + \beta \|\mathbf{S}\|_F^2 + \gamma \text{Tr}(\mathbf{L}^T (\mathbf{I}_N - \mathbf{S}) \mathbf{L}) \\ \text{s.t.} \quad & \mathbf{L}^T \mathbf{L} = \mathbf{I}_C, \sum_{j=1}^n S_{ij} = 1, S_{ij} \geq 0 \end{aligned} \quad (5)$$

Here, $\mathbf{K} \in \mathbb{R}^{N \times N}$ denotes the fused similarity matrix obtained from the previous kernel selection and fusion step. $\mathbf{S} \in \mathbb{R}^{N \times N}$ is the structure-preserving similarity matrix to be learned. $\mathbf{L} \in \mathbb{R}^{N \times C}$ is an auxiliary low-dimensional embedding matrix used to represent the samples in a C -dimensional space. N denotes the number of samples (e.g., patients), and C represents the number of final subtypes (i.e., cancer subtypes).

The first term in the objective function measures the distance between the fused kernel and the learned similarity matrix, reflecting the reconstruction error. The second term is a regularization term that encourages the learned similarity matrix to approach the unit simplex. If C is approximately equal to the number of subtypes, the low-rank structure of $\mathbf{S}$ aligns well with the desired cluster structure. The third term constrains $\mathbf{L}$ to be a valid embedding of $\mathbf{S}$ in a C -dimensional space by enforcing the orthonormality of $\mathbf{L}$. The matrix $(\mathbf{I}_N - \mathbf{S})$ essentially serves as a graph Laplacian, and its smallest C eigenvectors correspond to

the C smallest non-zero eigenvalues, which reflect the clustering structure of S .

We jointly optimize the two variables, S and L . The proposed optimization problem is non-convex. However, when any two variables are fixed, the subproblem for the remaining variable is convex. Thus, we can use an alternating optimization strategy to solve the optimization problem for these two variables efficiently.

- *Step 1 (Initialization):*

The similarity matrix S is initialized as:

$$S_0 = \max_{i,j} \left(\sum_{k=1}^m w_k \mathbf{D}_k(i,j) \right) - \sum_{k=1}^m w_k \mathbf{D}_k, \quad (6)$$

where L is initialized as the first C eigenvectors of the matrix $(I_N - S)$.

- *Step 2 (Fix L and update S):*

When minimizing the objective function with respect to the similarity matrix S , the optimization problem can be rewritten as:

$$\begin{aligned} \min_S \quad & \|K - S\|_F^2 + \beta \|S\|_F^2 - \gamma \text{Tr}(L^\top S L) \\ \text{s.t.} \quad & \sum_j S_{ij} = 1, \quad S_{ij} \geq 0 \end{aligned} \quad (7)$$

- *Step 3 (Fix S and update L):*

When minimizing the objective function with respect to the latent matrix L , the optimization problem becomes:

$$\begin{aligned} \max_L \quad & \text{Tr}(L^\top (S - I_N) L) \\ \text{s.t.} \quad & L^\top L = I_C \end{aligned} \quad (8)$$

According to matrix trace optimization theory [33], this objective is maximized when L consists of the top C eigenvectors of the matrix $(S - I_N)$, corresponding to its largest eigenvalues. Thus, L can be efficiently obtained using standard eigen decomposition routines in numerical linear algebra libraries.

- *Step 4 (Enhance similarity via diffusion):* After constructing the similarity matrix, to further mitigate the impact of noise and missing values inherent in omics data, we adopted the diffusion-based similarity enhancement mechanism from SIMLR [33]. This approach constructs a sparse transition matrix and iteratively refines the initial similarity through progressive diffusion updates. By preserving the original similarity structure while enhancing local details, the diffusion process significantly improves the robustness of cell-to-cell similarity. Moreover, it sharpens the boundaries between clusters during visualization and subtype identification, thereby strengthening the model's ability to capture underlying biological structures.

D. Cancer Subtype Identification

After obtaining the optimized sample similarity matrix S , we proceed to identify cancer subtypes for individuals. Let $y_{i(c)} \in \{0, 1\}$ denote the indicator variable for individual i belonging to the c -th cluster. Specifically, $y_{i(c)} = 1$ if individual i belongs to

the c -th subtype, and 0 otherwise. The clustering result can be represented as a partition matrix $Y = (Y_1; Y_2; \dots; Y_n)$.

Spectral clustering is adopted for its effectiveness in capturing the global structure of data, making it more suitable for high-dimensional and complex data clustering tasks compared to traditional K-means. By minimizing the following Ratio Cut objective function, this approach identifies the optimal clustering partition:

$$\min_Q \text{Tr}(Q^\top L Q) \quad \text{s.t.} \quad Q^\top Q = I. \quad (9)$$

where $Q = Y(YY^\top)^{-1/2}$, and $L = I - D^{-1/2} S D^{-1/2}$ is the normalized Laplacian matrix. Here, D is the degree matrix with diagonal entries $d_i = \sum_{j=1}^n S_{ij}$, representing the sum of similarities for node i .

IV. EXPERIMENTS

In this section, we evaluate the performance of our proposed algorithm for cancer subtype prediction through experiments on five multi-omics datasets.

A. Datasets and Evaluation

In this experiment, we applied the proposed algorithm to the TCGA dataset, which includes the following cancer types: Breast Invasive Carcinoma (TCGA-BRCA), Colon Adenocarcinoma (TCGA-COAD), Kidney Renal Clear Cell Carcinoma (TCGA-KIRC), Laryngeal Squamous Cell Carcinoma (TCGA-LUSC), and Glioblastoma Multiforme (TCGA-GBM), to identify subtypes associated with survival outcomes. For each cancer dataset, we analyzed three omics layers: mRNA expression (transcriptomics), DNA methylation (epigenomics), and miRNA expression (microRNA omics). Sample sizes ranged from 92 for the TCGA-COAD dataset to 215 for the TCGA-GBM dataset. Only patients with complete data across all three omics were included, while those missing any omics measurements were excluded. Table II summarizes the characteristics of the multi-omics datasets. We selected these three omics modalities because they are the most comprehensively profiled across TCGA cohorts, ensuring sufficient sample size for robust and reliable analysis. From a biological perspective, they capture complementary layers of tumor regulation: mRNA expression reflects downstream transcriptional programs; DNA methylation encodes epigenetic regulation that modulates gene activity; and miRNA expression provides insights into post-transcriptional control of mRNA stability and translation. Clinically, all three have been consistently associated with cancer initiation, progression, and prognosis, collectively representing a balanced integration of transcriptomic, epigenomic, and regulatory information. Therefore, this combination offers both broad molecular coverage and strong clinical relevance for uncovering biologically and clinically meaningful cancer subtypes.

We evaluated the performance of cancer subtype identification using three complementary metrics: (1) The concordance index (C-index) is a widely used metric for evaluating survival models [34]. It measures the proportion of all comparable subject pairs whose predicted survival times are correctly ordered. A

TABLE II
DETAIL OF FIVE DATASETS

Datasets	mRNA	Methylation	miRNA	Patients number	Survival time	Survival state
TCGA-BRCA	17814	23094	354	105	Int	Boolean
TCGA-COAD	17814	23088	312	92	Int	Boolean
TCGA-GBM	12042	1305	534	215	Int	Boolean
TCGA-KIRC	17899	24960	329	122	Int	Boolean
TCGA-LUSC	12042	23074	352	106	Int	Boolean

higher C-index corresponds to a better prediction performance (The value is between 0 and 1). (2)The silhouette score assesses clustering quality by balancing intra-cluster cohesion and inter-cluster separation. (3)The p -value from the Cox log-rank test [35] reflects the likelihood that the observed survival differences among subtypes occurred by chance rather than indicating genuine distinctions. The log-rank test statistic is defined as

$$\chi^2_{\log\text{-rank}} = \sum_{i=1}^c \frac{(O_i - E_i)^2}{E_i}, \quad (10)$$

where c denotes the number of subtype classes, O_i is the observed number of events in the i -th group, and E_i is the expected number of events, computed as

$$E_i = \sum_{j=1}^n \frac{R_{ij} D_j}{R_j}, \quad (11)$$

where n is the total number of event times, R_j represents the number of subjects at risk at the time of event j , R_{ij} is the number of subjects at risk from the i -th group at that time, and D_j is the total number of events (e.g., deaths) at time j .

B. Experimental Setup

In this study, we evaluated the performance of our proposed multi-omics cancer subtype identification method under two different experimental settings. The first setting follows the protocol described in [6], where the number of clusters is fixed across all methods for each dataset: 3 clusters for TCGA-GBM, TCGA-COAD, and TCGA-KIRC, 4 for TCGA-LUSC, and 5 for TCGA-BRCA. The second setting adopts a more flexible evaluation strategy, in which the number of clusters for each method is selected based on its optimal performance on each dataset, with the candidate range set between 3 and 6.

Under both scenarios, we systematically compared our method with a variety of advanced algorithms, covering multi-omics clustering, multi-view clustering, and multi-kernel clustering techniques. Specifically, we included advanced multi-omics clustering methods such as SNF [6] and CIMLR [10], state-of-the-art multi-view clustering algorithms such as AWMVC [36] and OMVCDR [37], and representative multi-kernel clustering approaches such as SimpleMKKM [38], MKKM-MR [39], and MKSSC-ERC [40]. In addition, we considered S2GC [41] based on graph structure learning and ONKC [42], a non-negative tensor factorization method. These methods cover a broad spectrum of clustering paradigms and provide a comprehensive benchmark to assess the effectiveness, robustness, and generalization capability of the proposed approach.

The parameter settings for the compared methods followed either the original papers or standard practices to ensure fair and consistent evaluation. For SNF [6], we searched the number of fusion iterations and neighbors over $\{20, 30, 40, 50\}$ and $\{10, 20, 30\}$, respectively. For S2GC [41], we adopted the default parameters as suggested in the original implementation ($\lambda = 1$, $\eta = 0.1$, $\gamma = 10$). For MKSSC-ERC [40], the parameters t and μ were set to $0.1n$ and $10^4 K$, respectively, and η was tuned over $\{2^{-8}, \dots, 2^8\}$. MKKM-MR [39] and ONKC [42] both involved tuning regularization parameters over the range $\{2^{-15}, \dots, 2^{15}\}$. CIMLR [10] involved optimizing kernel weights and tuning parameters β , γ , and ρ , with the number of clusters C ranging from 2 to 15. AWMVC [36] and OMVCDR [37] are based on matrix factorization, with latent space dimensionality determined by the number of clusters k and the number of embeddings m , typically set to 3. OMVCDR further includes a regularization parameter λ tuned over $\{2^{-5}, \dots, 2^5\}$. SimpleMKKM [38] was used with its default setting due to its simplicity and robustness.

To comprehensively evaluate the effectiveness of our method, we conducted detailed comparisons with several existing cancer subtyping approaches. To ensure fairness, we strictly adhered to the original implementations and parameter settings provided by the respective authors. For each method, no fewer than ten independent runs were performed, and the best results were selected for comparison. All experiments were implemented in MATLAB R2023b and executed on a Windows 11 platform equipped with an Intel(R) Core(TM) i9-14900HX CPU (2.20 GHz) and 16 GB of RAM.

C. Experimental Results

In this section, the parameters of all baseline algorithms were configured according to the recommendations provided in their original publications. Following the experimental protocol described in Section IV-B, we systematically evaluated the performance of the proposed method under two distinct experimental settings. The results for the first setting, where the number of clusters was fixed across all methods, are reported in Table III. In contrast, the performance of each method under the second setting, where the optimal number of clusters was selected from a predefined range, is provided in Appendix B in the supplemental material. Fig. 2 illustrates the results on the TCGA-BRCA dataset as a representative example, while the results for the remaining datasets are provided in Appendix B, available in the supplemental material.

1) Performance Analysis: From Table III, in the first experimental scenario (with a fixed number of clusters), our

TABLE III
P-VALUES FROM COX LOG-RANK TEST, C-INDEX, AND SILHOUETTE SCORES VALUES FOR CANCER SUBTYPE IDENTIFICATION METHODS ACROSS DATASETS.
BOLD NUMBERS INDICATE THE BEST RESULTS.

Datasets	Clusters	SNF	S2GC	MKSSC-ERC	SimpleMKKM	MKKM-MR	ONKC	CIMLR	AWMVC	OMVCDR	SIMKF
TCGA-BRCA(5)	P-value	1.1×10^{-3}	3.2×10^{-3}	8.7×10^{-6}	5.9×10^{-4}	9.5×10^{-9}	2.9×10^{-5}	4.0×10^{-4}	1.8×10^{-2}	3.8×10^{-3}	5.4×10^{-11}
	C-index	0.69	0.82	0.67	0.76	0.77	0.70	0.75	0.77	0.66	0.82
	Silhouette	0.75	0.48	0.51	0.39	0.39	0.32	0.76	0.38	0.42	0.94
TCGA-COAD(3)	P-value	3.6×10^{-2}	6.4×10^{-2}	6.3×10^{-4}	3.9×10^{-2}	4.7×10^{-3}	2.5×10^{-3}	4.4×10^{-1}	2.8×10^{-3}	5.3×10^{-4}	6.3×10^{-9}
	C-index	0.78	0.87	0.71	0.59	0.50	0.73	0.57	0.78	0.80	0.87
	Silhouette	0.73	0.22	0.51	0.77	0.89	0.39	0.81	0.47	0.32	0.92
TCGA-GBM(3)	P-value	5.7×10^{-4}	4.1×10^{-5}	3.4×10^{-4}	2.0×10^{-2}	6.2×10^{-5}	5.5×10^{-3}	1.4×10^{-4}	2.4×10^{-1}	4.9×10^{-3}	8.1×10^{-6}
	C-index	0.56	0.57	0.57	0.55	0.58	0.56	0.59	0.56	0.56	0.59
	Silhouette	0.47	0.67	0.70	0.35	0.43	0.38	0.95	0.48	0.48	0.99
TCGA-KIRC(3)	P-value	3.1×10^{-3}	2.6×10^{-1}	1.4×10^{-2}	3.1×10^{-1}	2.5×10^{-2}	1.1×10^{-1}	1.7×10^{-1}	5.6×10^{-5}	8.9×10^{-2}	2.3×10^{-4}
	C-index	0.58	0.56	0.57	0.54	0.61	0.64	0.60	0.63	0.61	0.66
	Silhouette	0.45	0.33	0.53	0.31	0.34	0.44	0.91	0.38	0.29	0.91
TCGA-LUSC(4)	P-value	1.8×10^{-2}	1.4×10^{-5}	1.9×10^{-4}	3.8×10^{-2}	2.0×10^{-4}	1.7×10^{-2}	1.7×10^{-1}	7.6×10^{-2}	4.3×10^{-3}	4.5×10^{-7}
	C-index	0.67	0.65	0.62	0.58	0.63	0.65	0.62	0.68	0.66	0.73
	Silhouette	0.39	0.35	0.77	0.41	0.44	0.47	0.71	0.54	0.46	0.93

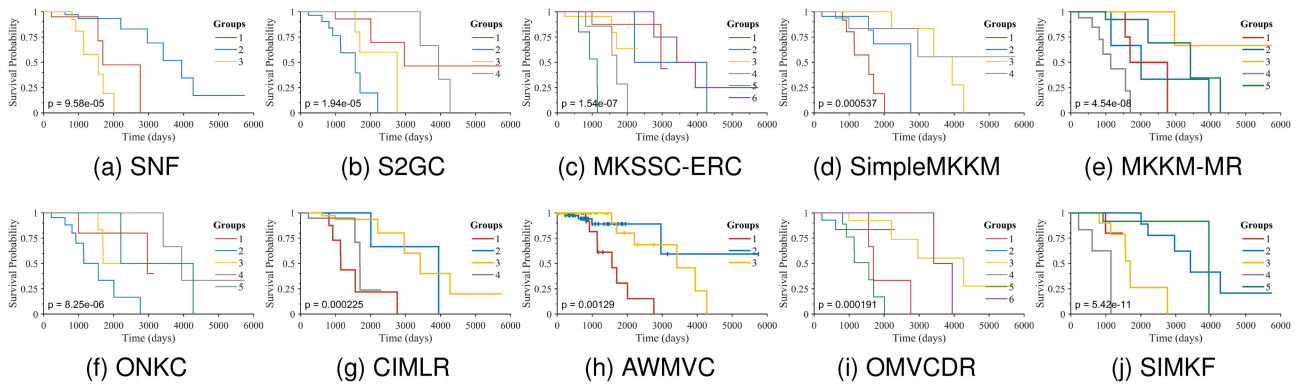


Fig. 2. The Kaplan Meier survival curves correspond to the best clustering results of each method in the TCGA-BRCA dataset.

method achieved significantly lower p -values on four out of five datasets, with the exception of TCGA-KIRC, indicating that the identified subtypes exhibit substantial survival differences and demonstrating the effectiveness of our approach. In addition, our method outperformed other approaches across all five datasets in terms of the concordance index (C-index) and silhouette score, reflecting both accurate survival ranking and stable, well-separated clustering. The comparatively weaker p -value for the TCGA-KIRC dataset can be attributed to the presence of at least one subtype with very few patients, making the p -values highly sensitive to small changes in clustering—a common issue for rare disease subtypes. Nevertheless, the C-index and silhouette score for TCGA-KIRC remained superior to other methods, indicating that our approach still effectively captures meaningful survival differences and produces stable clustering even in the presence of rare subtypes.

In the second experimental scenario, as shown in Fig. 2, our method yields more distinct survival curve separations on the TCGA-LUSC dataset, indicating improved subtype discrimination and more interpretable predictions of patient survival outcomes. In contrast, the alternative method exhibited weaker stratification, underscoring its limitations in identifying high-risk groups. Extended survival curves and additional experimental results for the remaining datasets are provided in Appendix B, available in the supplemental material. These

results further demonstrate our method's ability to accurately distinguish high-mortality subgroups—an essential criterion for evaluating the clinical utility of subtype prediction. The capacity to identify such high-risk groups holds substantial relevance for cancer diagnosis and treatment. Overall, our results validate not only the statistical advantage of the proposed method but also its strong potential for delivering accurate and clinically meaningful survival predictions.

Fig. 3 displays the visualization of clustering results obtained after dimensionality reduction of the integrated multi-omics data using the t-SNE method. Each point represents a sample, with the color indicating its predicted subtype. As shown in the figure, different subtypes are clearly separated in the two-dimensional space, with distinct boundaries and minimal overlap, demonstrating the strong discriminative ability of our method. Moreover, samples belonging to the same subtype are highly concentrated in the low-dimensional space, indicating good intra-cluster consistency and compactness. This clear inter-cluster separation and significant intra-cluster cohesion provide strong evidence for the effectiveness and robustness of the proposed method in multi-omics cancer subtype identification. Visualizations for other methods are provided in the Appendix C, available in the supplemental material.

2) *Effectiveness Verification*: In this subsection, we further evaluate the effectiveness of the proposed method, particularly

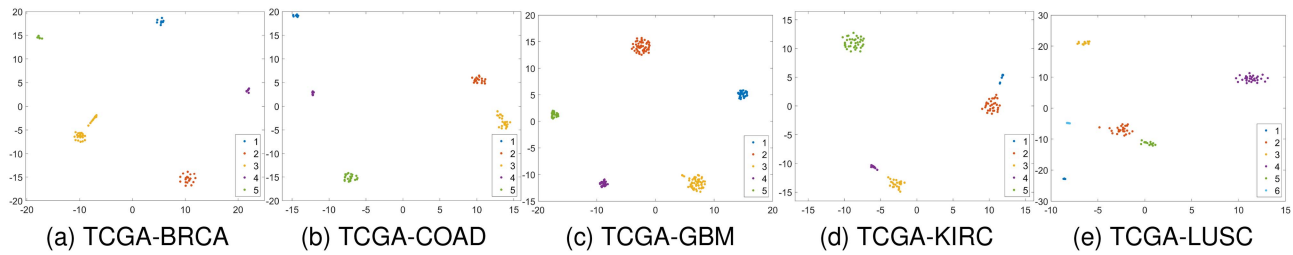


Fig. 3. Visualization of clustering results using a scatter plot.

TABLE IV

P-VALUES FROM COX LOG-RANK TEST FOR THE PROPOSED METHOD ON DIFFERENT INDIVIDUAL AND MULTI-OMIC DATASETS. BOLD NUMBERS INDICATE THE BEST RESULTS FOR EACH DATASET.

Dataset	mRNA	Methylation	miRNA	Integrated
TCGA-BRCA	2.9×10^{-4}	1.6×10^{-16}	2.0×10^{-4}	5.4×10^{-11}
TCGA-COAD	3.2×10^{-3}	1.7×10^{-2}	1.9×10^{-2}	2.9×10^{-9}
TCGA-KIRC	9.4×10^{-3}	2.1×10^{-4}	1.8×10^{-4}	3.8×10^{-7}
TCGA-LUSC	1.3×10^{-4}	1.8×10^{-11}	1.9×10^{-2}	7.3×10^{-10}
TCGA-GBM	5.1×10^{-2}	1.3×10^{-3}	5.6×10^{-3}	3.0×10^{-11}

TABLE V

P-VALUES FROM COX LOG-RANK TEST USING DIFFERENT KERNEL FUNCTIONS ACROSS DATASETS. BOLD NUMBERS INDICATE THE BEST RESULTS FOR EACH DATASET.

Dataset	Linear Kernel	Polynomial Kernel	Gaussian Kernel	Adopted Kernel
TCGA-BRCA	3.4×10^{-5}	6.5×10^{-6}	1.4×10^{-9}	5.4×10^{-11}
TCGA-COAD	7.2×10^{-4}	1.3×10^{-5}	2.5×10^{-6}	2.9×10^{-9}
TCGA-GBM	3.6×10^{-5}	9.1×10^{-6}	3.1×10^{-8}	3.0×10^{-11}
TCGA-KIRC	8.1×10^{-3}	1.7×10^{-3}	3.7×10^{-4}	3.8×10^{-7}
TCGA-LUSC	5.9×10^{-4}	4.5×10^{-3}	9.3×10^{-3}	7.3×10^{-10}

focusing on the kernel selection strategy guided by prior associations and diversity, as well as the adaptive kernel weight allocation mechanism based on the distribution characteristics of multi-omics data. Specifically, we compare the performance of the model integrating all three omics types with that of models trained using each omics data type individually. The comparison results are presented in Table IV. As shown in the table, the integration-based model consistently outperforms the single-omics models on most datasets. These results demonstrate that the proposed method can effectively integrate complementary information from different omics modalities, leading to significantly different survival profiles and improved clustering performance in most cases.

Furthermore, by comparing the results in Table III with those in Table IV, we observe that even when using only a single omics dataset, the proposed method still achieves superior performance compared to existing multi-omics integration approaches. This further confirms the robustness and wide applicability of the proposed strategy across different data scenarios. Notably, although the multi-omics integration model generally demonstrates stronger survival stratification ability across most datasets, in the TCGA-BRCA and TCGA-LUSC datasets the model constructed using only methylation (Methy) omics data yields lower log-rank p -values than the integration model. This indicates that, in these two specific datasets, methylation omics provides the most critical prognostic information, and its signal alone may be sufficient to distinguish patient survival risks. In contrast, when integrating multiple omics data, the presence of noise or redundant information from certain omics may slightly weaken the overall statistical significance.

To further validate the effectiveness of the selected kernels, we conducted comparative experiments across the five cancer datasets, applying several commonly used kernels—including linear, quadratic polynomial, radial basis function (RBF), and

the proposed scaled exponential similarity kernel with parameter perturbation to all three omics types in each dataset. The results, summarized in Table V, show that across the five cancer datasets, the proposed kernel consistently achieves the most significant p -values in survival analysis, outperforming the other classical kernels. This indicates that the proposed kernel not only effectively captures the heterogeneous information in multi-omics data but also better reflects the association between tumor subtypes and patient prognosis, thereby enhancing subtype identification after multi-omics integration. In comparison, linear and polynomial kernels have limitations in capturing complex nonlinear relationships, while the traditional RBF kernel, although capable of modeling certain nonlinearities, does not fully exploit the complementary information across omics in integration, resulting in slightly inferior performance.

Therefore, these observations suggest that in multi-omics integration, attention should be paid to the weight assignment across different omics, and appropriately emphasizing omics types that carry richer prognostic information could further improve the performance of the integration model. At the same time, this also reflects that multi-omics fusion is not absolutely superior to single-omics models in all scenarios, as its effectiveness depends on both the data quality and the disease relevance of each omics type. Collectively, the results provide strong evidence for the advantages of the proposed kernel selection and weighting strategies in multi-omics integration and survival-related subtype identification.

3) *Clinical Analysis on Breast Cancer Data (TCGA-BRCA):* As shown in Fig. 5, the optimal numbers of clusters predicted on the TCGA-BRCA dataset using the separation cost function described earlier are 3 and 5, corresponding to two local minima of the cost curve. These results indicate that both clusterings capture biologically meaningful structures. However, further survival analysis reveals that when the number of clusters is set to 5, the resulting subtypes exhibit more pronounced divergence in

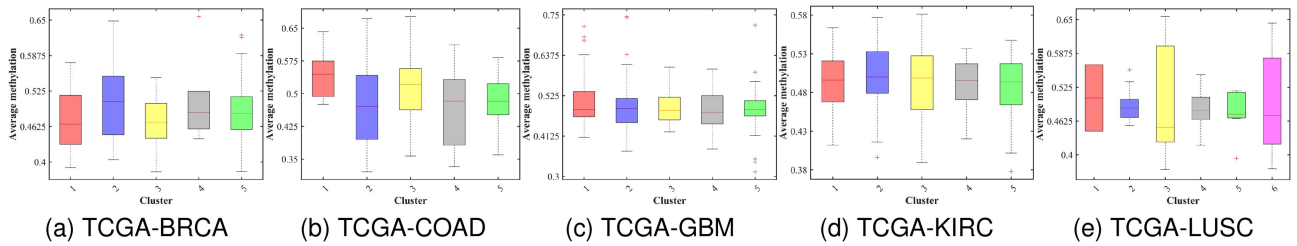


Fig. 4. Boxplot of Average Methylation β -Values for Each Subcluster. β -values indicate DNA methylation levels at CpG sites, ranging from 0 (unmethylated) to 1 (fully methylated).

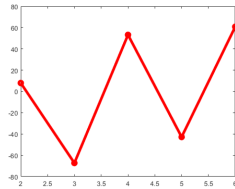


Fig. 5. Estimate the best number of clusters for the TCGA-BRCA dataset.

TABLE VI
ABLATION STUDY

Dataset	R_SKS	R_MMD	SIMKF
TCGA-BRCA	2.2×10^{-3}	3.5×10^{-9}	5.4×10^{-11}
TCGA-COAD	4.2×10^{-2}	1.1×10^{-5}	2.9×10^{-9}
TCGA-GBM	4.3×10^{-4}	1.0×10^{-7}	3.0×10^{-11}
TCGA-KIRC	1.1×10^{-3}	2.2×10^{-4}	3.8×10^{-7}
TCGA-LUSC	3.1×10^{-5}	1.6×10^{-6}	7.3×10^{-10}

survival profiles, with a smaller log-rank p -value. This suggests stronger survival heterogeneity among the identified subtypes.

This observation indicates that our method is capable of discovering subtype stratifications that are more closely associated with clinical prognosis. Furthermore, it demonstrates the method's sensitivity in capturing latent biological differences. Compared to the conventional 3-subtype partitioning—which aligns with known subtypes such as Luminal A, Luminal B, HER2-enriched, and Basal-like—the 5-cluster solution produced by our algorithm enables a finer-grained decomposition, uncovering previously indistinguishable subgroups with distinct clinical or biological characteristics.

Therefore, the results substantiate the robustness and expressiveness of our approach, highlighting its effectiveness and practical value in identifying biologically and clinically meaningful subtypes from complex multi-omic data.

Breast cancer is typically classified into five molecular subtypes based on four key biomarkers: estrogen receptor (ER), progesterone receptor (PR), human epidermal growth factor receptor 2 (HER2), and the Ki-67 proliferation index [43], [44], [45]. These subtypes include Luminal A, Luminal B, HER2-positive (HR-negative), HER2-positive (HR-positive), and triple-negative breast cancer (TNBC). In this study, the proposed SIMKF method was applied to a breast cancer dataset comprising 105 samples (TCGA-BRCA), successfully identifying five distinct clusters that exhibited significant differences in both overall survival (Fig. 2(j)) and DNA methylation profiles (Fig. 4(a)). Survival analysis shows a non-linear relationship between methylation levels and breast cancer prognosis. Cluster 2 exhibits the best survival outcome, characterized by high and stable methylation, suggesting that moderate hypermethylation may help maintain proper gene regulation and inhibit tumor progression. Cluster 4 has the worst prognosis despite only moderately low methylation, indicating that factors beyond methylation, such as genetic mutations or chromosomal abnormalities,

may drive its poor outcomes. Cluster 1 also displays poor survival, where slightly reduced methylation relative to Cluster 2 may already disrupt tumor-suppressive regulation. Cluster 3 has the lowest methylation but does not show any survival advantage, implying that hypomethylation may promote oncogenic activity rather than improve prognosis. Cluster 5 initially shows favorable survival but declines later, possibly due to limited follow-up or small sample size; its moderately high and stable methylation suggests short-term benefit with uncertain long-term trends. Overall, both excessive and insufficient methylation can be associated with unfavorable survival, highlighting the complex and non-monotonic role of methylation in breast cancer progression.

4) *Ablation Study:* To further validate the effectiveness of the proposed method, we conducted ablation experiments. Specifically, we designed two variant models: R-SKS, which removes the Survival-Guided Kernel Selection (SKS) component to evaluate the contribution of survival information in selecting informative kernels, and R-MMD, which removes the Distribution-Aware Kernel Fusion (MMD-based fusion) component to assess the role of distribution-aware weighting in integrating complementary kernel information.

The results of the ablation study are summarized in Table VI. It can be observed that removing the survival-guided kernel selection component (R-SKS) leads to a noticeable performance drop, particularly reflected in higher log-rank p -values for survival stratification. This indicates that survival-guided kernel selection is crucial for filtering informative kernels and reducing interference from redundant or irrelevant kernels. Similarly, removing the MMD-based kernel fusion component (R-MMD) also results in decreased performance, highlighting the importance of distribution-aware weighted fusion in effectively integrating complementary kernel information.

Overall, these ablation results demonstrate that both the survival-guided kernel selection and the MMD-based kernel

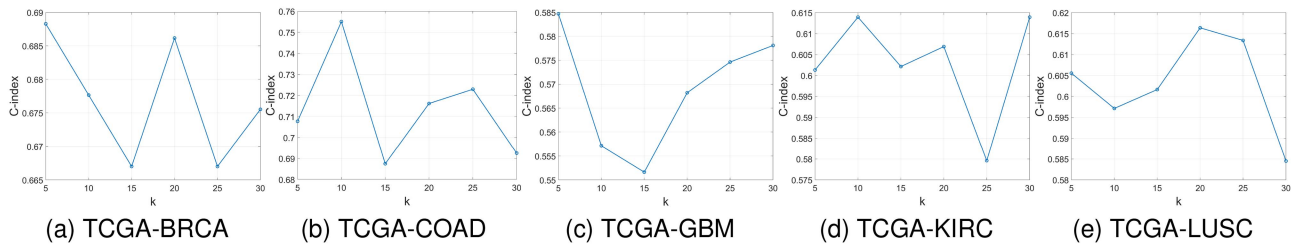
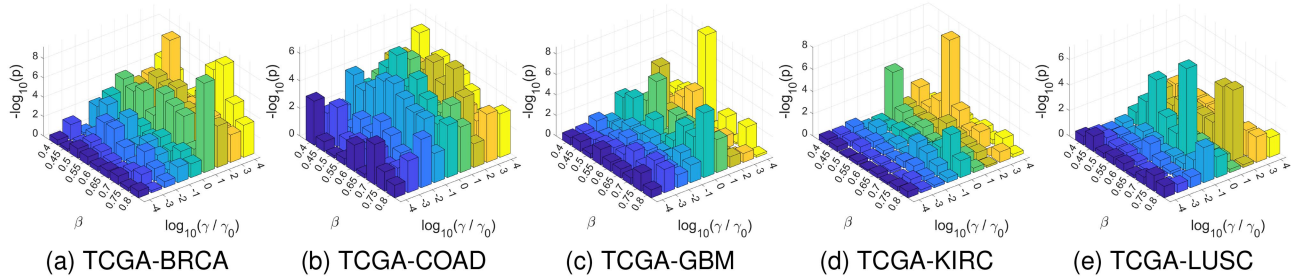
Fig. 6. Sensitivity of clustering performance to the number of nearest neighbors k on five datasets.

Fig. 7. Sensitivity of clustering performance to the two-hyper-parameter space.

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT FUSION METHODS

Dataset	SIMKF_AVG	SIMKF_Iter	SIMKF_DST	SIMKF
TCGA-BRCA	8.3×10^{-4}	3.5×10^{-9}	7.9×10^{-8}	5.4×10^{-11}
TCGA-COAD	1.8×10^{-5}	1.1×10^{-5}	4.6×10^{-7}	2.9×10^{-9}
TCGA-GBM	1.0×10^{-5}	1.0×10^{-7}	9.8×10^{-9}	3.0×10^{-11}
TCGA-KIRC	2.2×10^{-2}	2.2×10^{-4}	4.7×10^{-5}	3.8×10^{-7}
TCGA-LUSC	1.6×10^{-3}	1.6×10^{-5}	2.8×10^{-8}	7.3×10^{-10}

fusion are essential components that enable SIMKF to robustly and accurately identify clinically meaningful cancer subtypes.

SIMKF learns weights for each omics dataset during the Multi-Kernel Similarity Fusion process, rather than assigning equal importance to all datasets. To further validate the effectiveness of our weight allocation strategy, we designed four different fusion methods: SIMKF-AVG, which uses a simple average weight allocation strategy, assigning equal weights to all omics data; SIMKF-Iter, which draws inspiration from the SimpleMKKM and CIMLR approaches by treating weight allocation as an optimization term in the iterative process. In each iteration, the algorithm adaptively adjusts the weights based on the features of each omics dataset, with the optimization goal of improving the clustering performance of the fused kernel; SIMKF-DST, which applies Dempster-Shafer Theory (DST) [46] for weight allocation. DST allows the fusion process to account for the evidence provided by each omics dataset, assigning weights based on the uncertainty and information overlap between the datasets; and SIMKF, our proposed method. The results are summarized in Table VII, it can be observed that the SIMKF method outperforms SIMKF-AVG, SIMKF-Iter, and SIMKF-DST in terms of log-rank p-values for survival stratification and subtype stability.

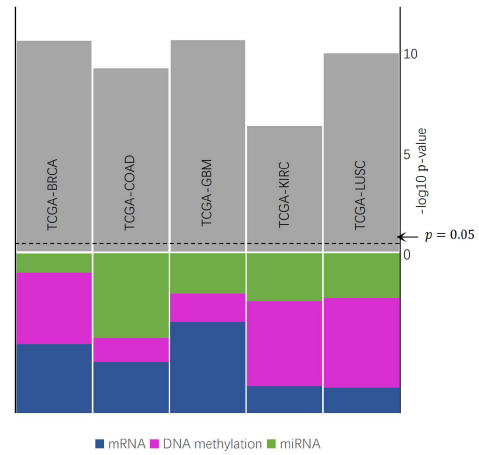


Fig. 8. Distribution of Kernel Weights for Different Omics Data Types Across Various Cancer Types.

We note that the contribution of each data type, measured as the fraction of total kernel weight based on the kernel contributions of that data type, varies significantly across different cancers (Fig. 8), the gray bars represent the level of significant differences in patient survival between clusters. DNA methylation and mRNA contribute roughly 30% -50% of the kernel weights across most cancer types, while miRNA weights vary considerably. We observe that these kernel weights are associated with the C/M classification of cancers [47]. Compared to C-type cancers, such as BRCA and LUSC, M-type cancers, such as COAD, GBM, and KIRC, exhibit higher contributions from miRNA. Therefore, SIMKF can help us understand which data types provide the most informative value for cancer subtyping

across different cancer types. The contribution distributions of the four types are provided in Appendix E, available in the supplemental material.

5) *Parameter Sensitivity Analysis*: Our method involves four key parameters: the number of nearest neighbors, two hyperparameters (β and γ), and the number of selected kernels C . To investigate the impact of these parameters on model performance, we conducted a parameter sensitivity analysis. Figs. 6 and 7 illustrate how variations in each parameter affect the performance of our model.

Our method involves four key parameters: the number of nearest neighbors, two hyperparameters (β and γ), and the number of selected kernels C . To investigate the impact of these parameters on model performance, we conducted a parameter sensitivity analysis. Figs. 6 and 7 illustrate how variations in each parameter affect the performance of our model.

In the experiments, we fixed $\beta = 0.8$ and $\gamma = 0.1$ to evaluate the influence of the nearest neighbor number k , and fixed $k = 20$ to assess the effects of β and γ . Specifically, γ serves as a regularization term to prevent each cell from having only a few non-zero similarities with others. The default value γ_0 is estimated in a data-driven manner based on the distance gap.

From the results, Fig. 6 shows that the model is not particularly sensitive to the choice of k . Therefore, we set the default value of k as the sample size of the dataset divided by 10. Fig. 7 demonstrates that the model performance is relatively insensitive to changes in β , while it is more sensitive to γ . In particular, when $\log_{10}(\gamma/\gamma_0)$ becomes too large or too small, the model performance exhibits noticeable fluctuations.

These observations suggest that our method is generally robust to the choice of parameters within a reasonable range, which further highlights its practical applicability across different datasets.

V. CONCLUSION

This study proposes a survival-informed multi-omics kernel fusion method (SIMKF) aimed at addressing key challenges in cancer subtyping, including multi-omics data heterogeneity, kernel selection, and weight assignment. Unlike existing methods that often overlook kernel quality, modality importance, and clinical phenotype relevance, SIMKF introduces a kernel selection strategy that integrates phenotype correlations—such as survival information—with kernel diversity, effectively identifying a representative and complementary subset of kernels. Additionally, SIMKF employs an unsupervised adaptive kernel weighting mechanism based on Maximum Mean Discrepancy (MMD), which avoids complex iterative optimization and enhances the discriminative power and clinical interpretability of the fused kernel matrix.

The proposed method demonstrates superior performance in subtype identification across multiple real-world multi-omics cancer datasets. The identified subtypes show significant associations with patient survival, underscoring their clear clinical prognostic value. Further biological analyses reveal that these subtypes uncover key molecular mechanisms and potential biomarkers. For example, clinical analysis of the breast cancer dataset (TCGA-BRCA) aligns well with established clinical

findings, further validating the effectiveness of SIMKF in identifying clinically relevant cancer subtypes. Future work will integrate gene expression profiles, somatic mutation data, and clinical subtype information to further explore the molecular mechanisms and clinical significance of each subtype. Special emphasis will be placed on the functional roles and signaling pathways associated with differentially methylated genes, aiming to advance precision subtyping and personalized treatment strategies in breast cancer.

REFERENCES

- [1] M. Roger et al., “Comprehensive genomic characterization defines human glioblastoma genes and core pathways,” *Nature*, vol. 455, no. 7216, pp. 1061–1068, 2008.
- [2] Y. Hasin, M. Seldin, and A. Lusi, “Multi-omics approaches to disease,” *Genome Biol.*, vol. 18, pp. 1–15, 2017.
- [3] T. M. Santiago-Rodriguez and E. B. Hollister, “Multi ‘omic data integration: A review of concepts, considerations, and approaches,” *Seminars Perinatology*, vol. 45, no. 6, 2021, Art. no. 151456.
- [4] D. Wu et al., “Fast dimension reduction and integrative clustering of multi-omics data using low-rank approximation: Application to cancer molecular classification,” *BMC Genomic.*, vol. 16, pp. 1–10, 2015.
- [5] T. Nguyen, R. Tagett, D. Diaz, and S. Draghici, “A novel approach for data integration and disease subtyping,” *Genome Res.*, vol. 27, no. 12, pp. 2025–2039, 2017.
- [6] B. Wang et al., “Similarity network fusion for aggregating data types on a genomic scale,” *Nature Methods*, vol. 11, no. 3, pp. 333–337, 2014.
- [7] E. F. Lock, K. A. Hoadley, J. S. Marron, and A. B. Nobel, “Joint and individual variation explained (JIVE) for integrated analysis of multiple data types,” *Ann. Appl. Statist.*, vol. 7, no. 1, 2013, Art. no. 523.
- [8] R. Shen, A. B. Olshen, and M. Ladanyi, “Integrative clustering of multiple genomic data types using a joint latent variable model with application to breast and lung cancer subtype analysis,” *Bioinformatics*, vol. 25, no. 22, pp. 2906–2912, 2009.
- [9] H. Yang, S. Li, H. Cao, C. Zhang, and Y. Cui, “Predicting disease trait with genomic data: A composite kernel approach,” *Brief. Bioinf.*, vol. 18, no. 4, pp. 591–601, 2017.
- [10] D. Ramazzotti, A. Lal, B. Wang, S. Batzoglou, and A. Sidow, “Multi-omic tumor data reveal diversity of molecular mechanisms that correlate with survival,” *Nature Commun.*, vol. 9, no. 1, pp. 1–14, 2018.
- [11] W. Zhang et al., “Integrating genomic, epigenomic, and transcriptomic features reveals modular signatures underlying poor prognosis in ovarian cancer,” *Cell Rep.*, vol. 4, no. 3, pp. 542–553, 2013.
- [12] D. Sun, M. Wang, and A. Li, “A multimodal deep neural network for human breast cancer prognosis prediction by integrating multi-dimensional data,” *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 16, no. 3, pp. 841–850, May/Jun. 2019.
- [13] H. Wang, H. Zheng, J. Wang, C. Wang, and F.-X. Wu, “Integrating omics data with a multiplex network-based approach for the identification of cancer subtypes,” *IEEE Trans. Nanobiosci.*, vol. 15, no. 4, pp. 335–342, Jun. 2016.
- [14] K. A. Hoadley et al., “Multiplatform analysis of 12 cancer types reveals molecular classification within and across tissues of origin,” *Cell*, vol. 158, no. 4, pp. 929–944, 2014.
- [15] S. Sun, W. Dong, and Q. Liu, “Multi-view representation learning with deep Gaussian processes,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 12, pp. 4453–4468, Dec. 2021.
- [16] S. Sun, M. Yu, J. Shawe-Taylor, and L. Mao, “Stability-based PAC-Bayes analysis for multi-view learning algorithms,” *Inf. Fusion*, vol. 86, pp. 76–92, 2022.
- [17] J. Feng, L. Jiang, S. Li, J. Tang, and L. Wen, “Multi-omics data fusion via a joint kernel learning model for cancer subtype discovery and essential gene identification,” *Front. Genet.*, vol. 12, 2021, Art. no. 647141.
- [18] H. Yang, H. Cao, T. He, T. Wang, and Y. Cui, “Multilevel heterogeneous omics data integration with kernel fusion,” *Brief. Bioinf.*, vol. 21, no. 1, pp. 156–170, 2020.
- [19] C. R. John, D. Watson, M. R. Barnes, C. Pitzalis, and M. J. Lewis, “Spectrum: Fast density-aware spectral clustering for single and multi-omic data,” *Bioinformatics*, vol. 36, no. 4, pp. 1159–1166, 2020.
- [20] J. Chen, W. Rong, G. Tao, and H. Cai, “Similarity fusion via exploiting high order proximity for cancer subtyping,” *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 20, no. 1, pp. 658–667, Jan./Feb. 2023.

- [21] Y. Wei et al., "Cancer subtyping with heterogeneous multi-omics data via hierarchical multi-kernel learning," *Brief. Bioinf.*, vol. 24, no. 1, 2023, Art. no. bbac488.
- [22] V. Crippa et al., "Characterization of cancer subtypes associated with clinical outcomes by multi-omics integrative clustering," *Comput. Biol. Med.*, vol. 162, 2023, Art. no. 107064.
- [23] K. A. Hoadley et al., "Cell-of-origin patterns dominate the molecular classification of 10,000 tumors from 33 types of cancer," *Cell*, vol. 173, no. 2, pp. 291–304, 2018.
- [24] V. Thorsson et al., "The immune landscape of cancer," *Immunity*, vol. 48, no. 4, pp. 812–830, 2018.
- [25] T. M. Malta et al., "Machine learning identifies stemness features associated with oncogenic dedifferentiation," *Cell*, vol. 173, no. 2, pp. 338–354, 2018.
- [26] F. Sanchez-Vega et al., "Oncogenic signaling pathways in the cancer genome atlas," *Cell*, vol. 173, no. 2, pp. 321–337, 2018.
- [27] A. Arora, A. B. Olshen, V. E. Seshan, and R. Shen, "Pan-cancer identification of clinically relevant genomic subtypes using outcome-weighted integrative clustering," *Genome Med.*, vol. 12, pp. 1–13, 2020.
- [28] H. Cao et al., "wmkl: Multi-omics data integration enables novel cancer subtype identification via weight-boosted multi-kernel learning," *Brit. J. Cancer*, vol. 130, no. 6, pp. 1001–1012, 2024.
- [29] D. Huang, C.-D. Wang, J.-H. Lai, and C.-K. Kwoh, "Toward multidiversified ensemble clustering of high-dimensional data: From subspaces to metrics and beyond," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 12231–12244, Nov. 2022.
- [30] X. Ding, J. Liu, F. Yang, and J. Cao, "Random radial basis function kernel-based support vector machine," *J. Franklin Inst.*, vol. 358, no. 18, pp. 10121–10140, 2021.
- [31] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," *J. Amer. Soc. Inf. Sci.*, vol. 41, no. 6, pp. 391–407, 1990.
- [32] K. M. Borgwardt, A. Gretton, M. J. Rasch, H.-P. Kriegel, B. Schölkopf, and A. J. Smola, "Integrating structured biological data by kernel maximum mean discrepancy," *Bioinformatics*, vol. 22, no. 14, pp. e49–e57, 2006.
- [33] B. Wang, J. Zhu, E. Pierson, D. Ramazzotti, and S. Batzoglou, "Visualization and analysis of single-cell RNA-seq data by kernel-based similarity learning," *Nature Methods*, vol. 14, no. 4, pp. 414–416, 2017.
- [34] H. Steck, B. Krishnapuram, C. Dehing-Oberije, P. Lambin, and V. C. Raykar, "On ranking in survival analysis: Bounds on the concordance index," in *Proc. Adv. Neural Inf. Process. Syst.*, 2007, vol. 20, pp. 1–8.
- [35] D. W. Hosmer, S. Lemeshow, and S. May, *Applied Survival Analysis: Regression Modeling of Time-to-Event Data*. Hoboken, NJ, USA: Wiley, 2008.
- [36] X. Wan et al., "Auto-weighted multi-view clustering for large-scale data," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 8, pp. 10078–10086.
- [37] X. Wan et al., "One-step multi-view clustering with diverse representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 3, pp. 5774–5786, Mar. 2025.
- [38] X. Liu, "SimpleMKKM: Simple multiple kernel k-means," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5174–5186, Apr. 2023.
- [39] X. Liu, Y. Dou, J. Yin, L. Wang, and E. Zhu, "Multiple kernel k-means clustering with matrix-induced regularization," in *Proc. AAAI Conf. Artif. Intell.*, 2016, vol. 30, no. 1, pp. 1888–1894.
- [40] Z. Xu et al., "Multi-kernel subspace stable clustering with exact rank constraints," *Inf. Fusion*, vol. 110, 2024, Art. no. 102488.
- [41] C. Liu et al., "Supervised graph clustering for cancer subtyping based on survival analysis and integration of multi-omic tumor data," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 19, no. 2, pp. 1193–1202, Mar./Apr. 2022.
- [42] X. Liu et al., "Optimal neighborhood kernel clustering with multiple kernels," in *Proc. AAAI Conf. Artif. Intell.*, 2017, vol. 31, no. 1, pp. 2266–2272.
- [43] C. M. Perou et al., "Molecular portraits of human breast tumours," *Nature*, vol. 406, no. 6797, pp. 747–752, 2000.
- [44] A. Goldhirsch et al., "Personalizing the treatment of women with early breast cancer: Highlights of the St Gallen International Expert Consensus on the primary therapy of early breast cancer 2013," *Ann. Oncol.*, vol. 24, no. 9, pp. 2206–2223, 2013.
- [45] M. Leeha, K. Kanokwiroon, S. Laohawiriyakamol, and P. Thongsuksai, "Immunohistochemistry-based molecular subtyping of triple-negative breast cancer and its prognostic significance," *Pathol. Oncol. Res.*, vol. 29, 2023, Art. no. 1611162.
- [46] A. P. Dempster, "Upper and lower probabilities induced by a multivalued mapping," in *Classic Works of the Dempster-Shafer Theory of Belief Functions*. Berlin, Germany: Springer, 2008, pp. 57–72.
- [47] G. Ciriello, M. L. Miller, B. A. Aksoy, Y. Senbabaoglu, N. Schultz, and C. Sander, "Emerging landscape of oncogenic signatures across human cancers," *Nature Genet.*, vol. 45, no. 10, pp. 1127–1133, 2013.



Xiaojian Ding received the PhD degree in computer science and technology from Xi'an Jiaotong University, China, in 2011. From 2009 to 2010, he was a visiting scholar with the Nanyang Technological University, Singapore. He was with the Laboratory of Science and Technology on Information Systems Engineering, Nanjing, China, and was with Huawei Technologies Company, Ltd., Nanjing, China. He joined the Nanjing University of Finance and Economics. He is currently a senior engineer with the Nanjing University of Finance and Economics. His

research interests include multi-view learning, and multi-modal learning. He has authored publications in conferences, such as NeurIPS, AAAI, IJCAI, and ACM MM, and is also a PC member for conferences including ICML, ICLR, NeurIPS, AISTATS, and AAAI. He is also an associate editor for *Pattern Analysis and Applications*.



Pengcheng Shi is currently working toward the MS degree in software engineering with the Nanjing University of Finance and Economics, Nanjing, China. His research interests include multi-omics data integration, cancer subtyping, and multi-kernel clustering.



Xin Wang is currently working toward the MS degree in computer science and technology with the Nanjing University of Finance and Economics, Nanjing, China. His research interests include multi-kernel clustering and multi-omics data integration.



Hui Cao received the master's degree from the School of Computer Science and Engineering, Xi'an University of Technology, China, in 2017. She is currently with PetroChina Changqing Oilfield Company Limited. Her research interests mainly include machine learning, deep learning, and large model research.